

GARMIN®

AVIATION TRAINING

EFIS System Design

AEA 2026

Garmin Aviation Training is intended to provide the best practices for the installation, use and maintenance of Garmin Aviation products. It is incumbent on technicians and pilots to ensure that the instructions contained in the most recent versions of Garmin product manuals, bulletins and notifications are followed.

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- ✈️ This course will focus on the various data bussing systems Garmin uses for integration of our Federated EFIS systems to work like our Flight Deck Systems.
- ✈️ At the end of this lesson the learner should understand the data bussing of the Garmin EFIS systems and the implications of decisions made

✈ The G3X system runs on can bus

✈ The can bus must be wired properly to fully function



Controller Area Network (Bosch)

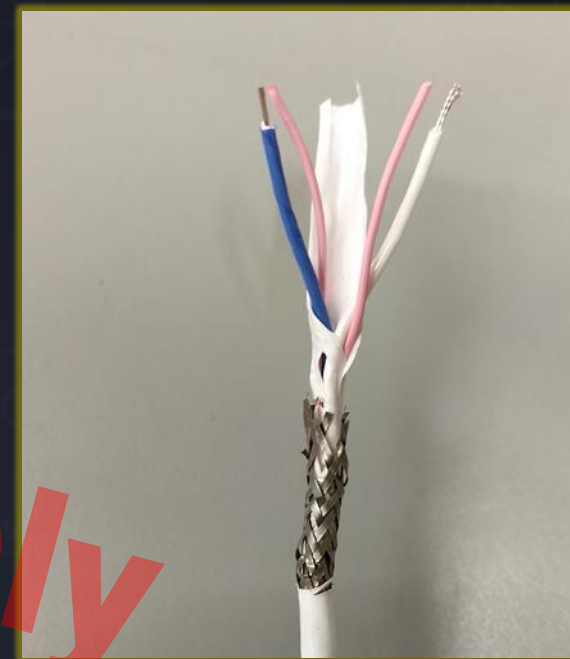
- ✈ Primary digital interface used to exchange data between G3X, G5, and GFC 500
- ✈ Widely used in aviation, automotive, and industrial applications
 - ✈ Simple wiring
 - ✈ Fast data stream
 - ✈ High reliability



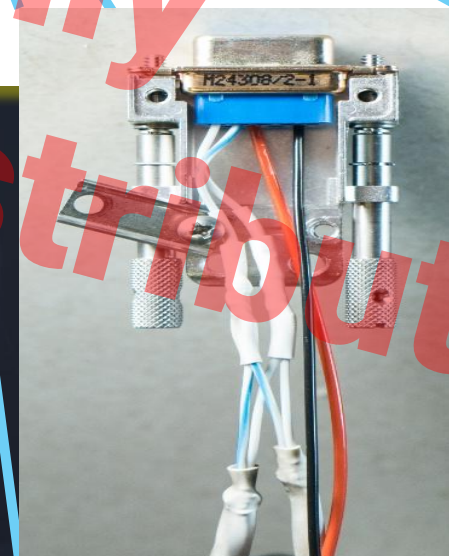
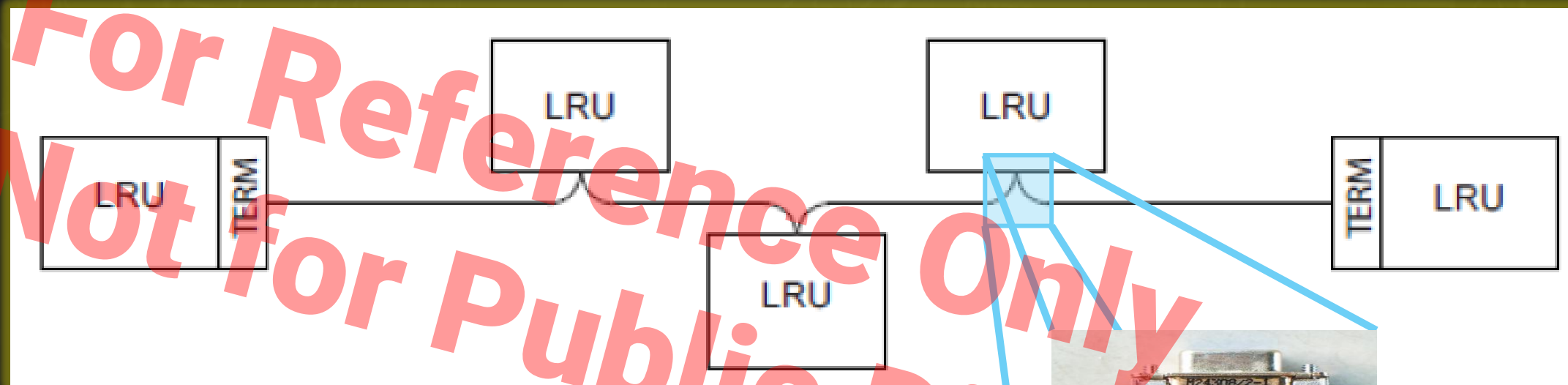
120 Ohm CAN Bus wire

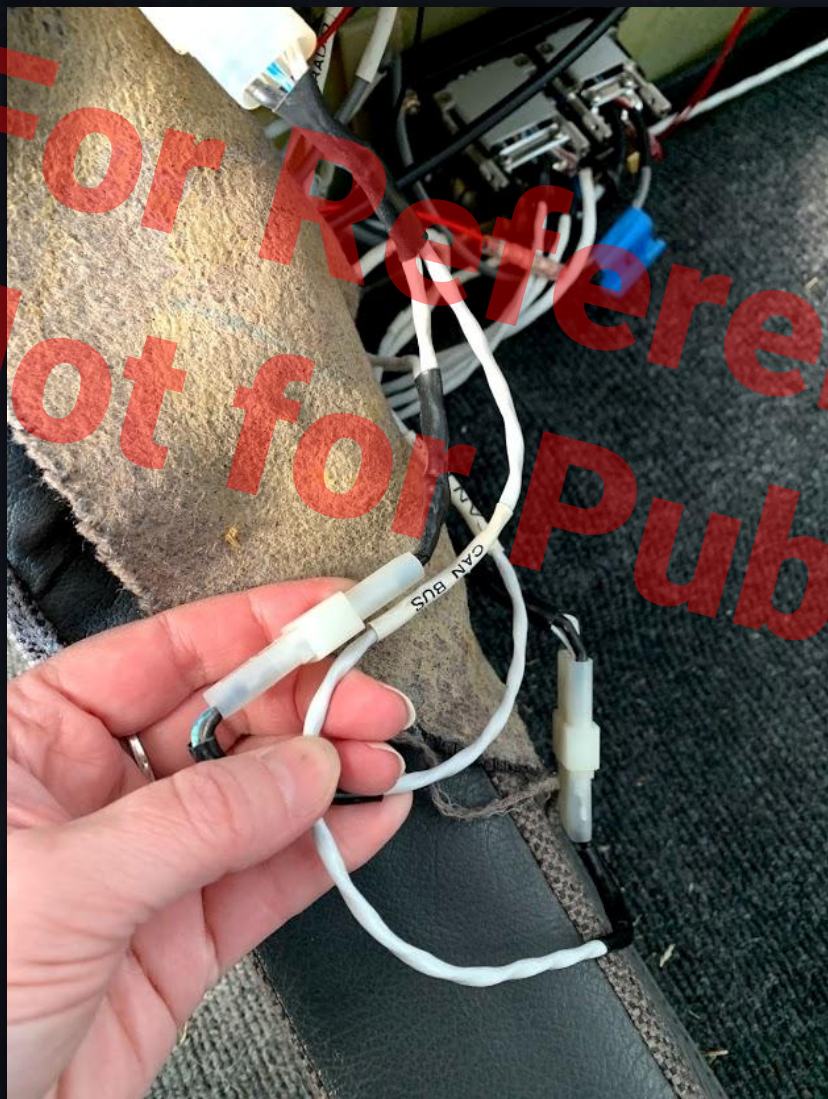
✈️ Two suppliers:

- ✈️ Carlisle IT P/N CAN24TST120(CIT)
- ✈️ GigaFlight Connectivity P/N GF120-24CANB-1

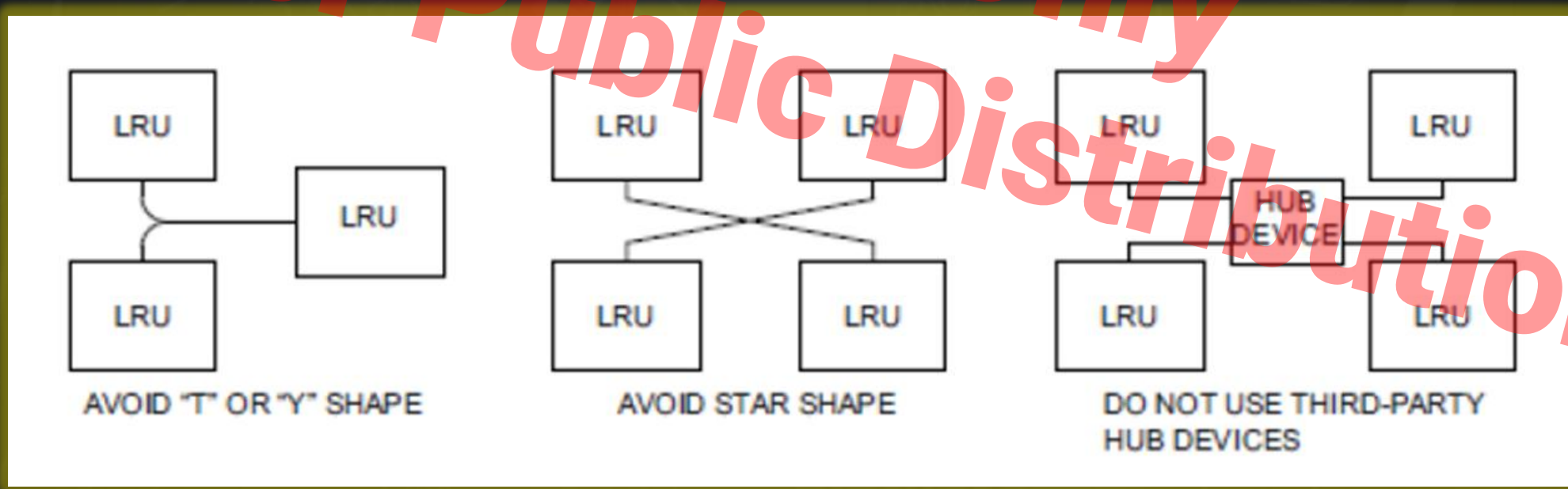


- ✈ Each LRU is daisy-chained into the bus





- ✈️ Bus string needs to be linear
- ✈️ Criss-crossing and back-tracking can cause signal interference and reflections in the data path.
- ✈️ T or Y shape splices will cause data packets to collide or be isolated, and increase error rates.

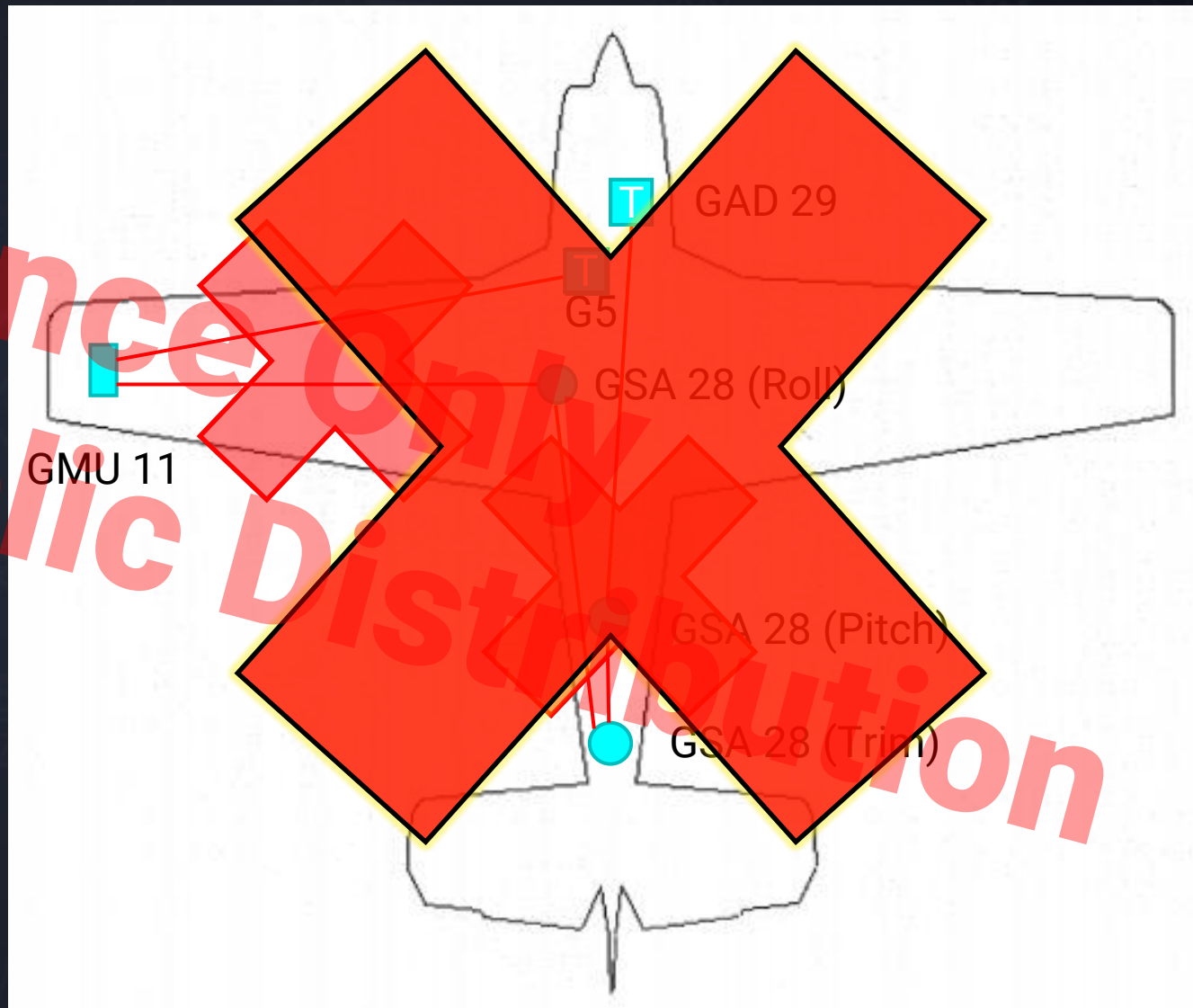


✈️ How should you arrange LRUs on the bus?

✈️ GMU placement varies from airframe to airframe due to differences in equipment, wiring, airframe construction

✈️ Which LRUs should we use to terminate the bus?

✈️ Terminate at the instrument panel??



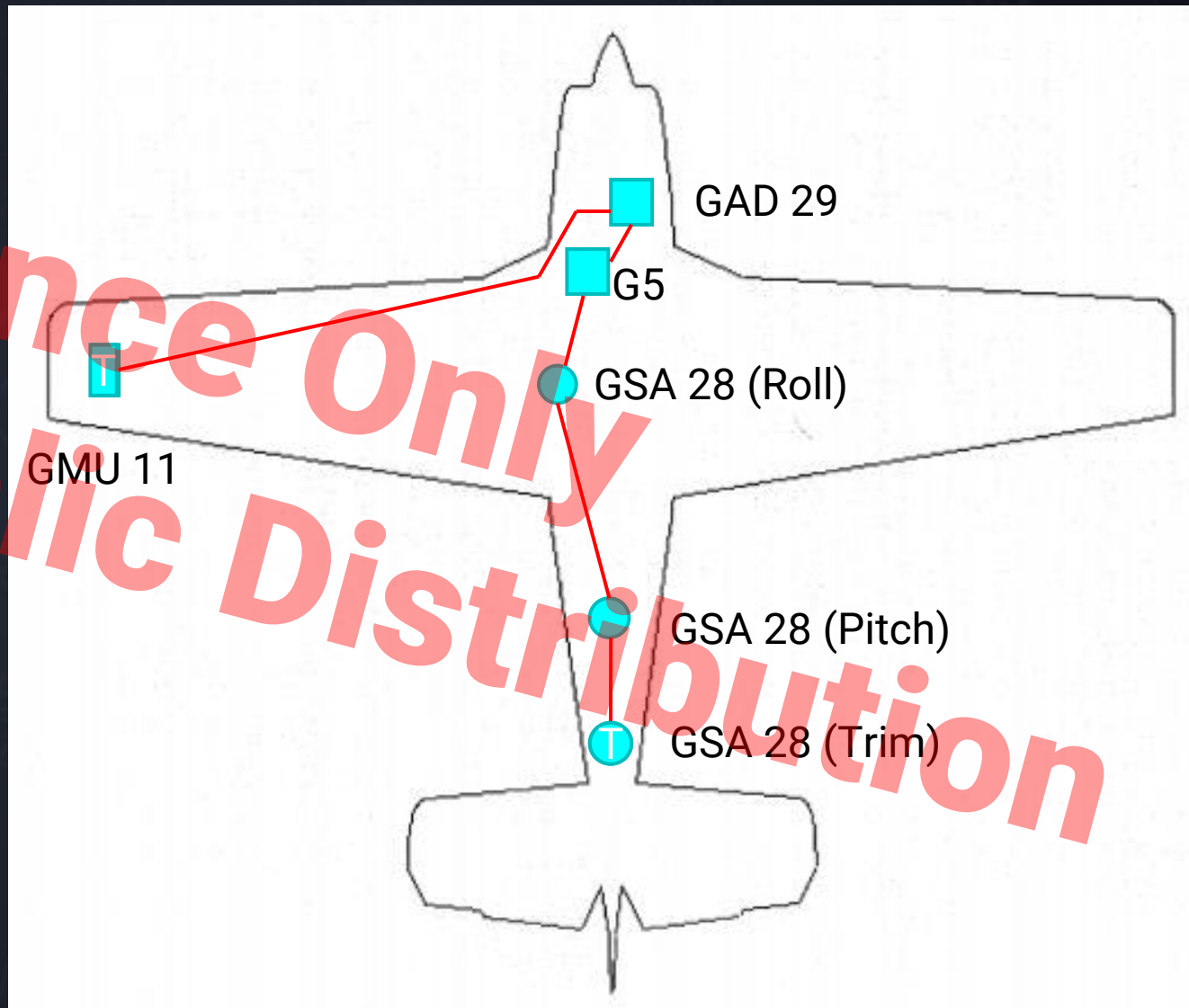
✈️ LRUs furthest out should be the termination:

✈️ GMU 11

✈️ GSA 28

✈️ Cuts length of bus down

✈️ Simplifies route for the data and eliminates potential interference sources



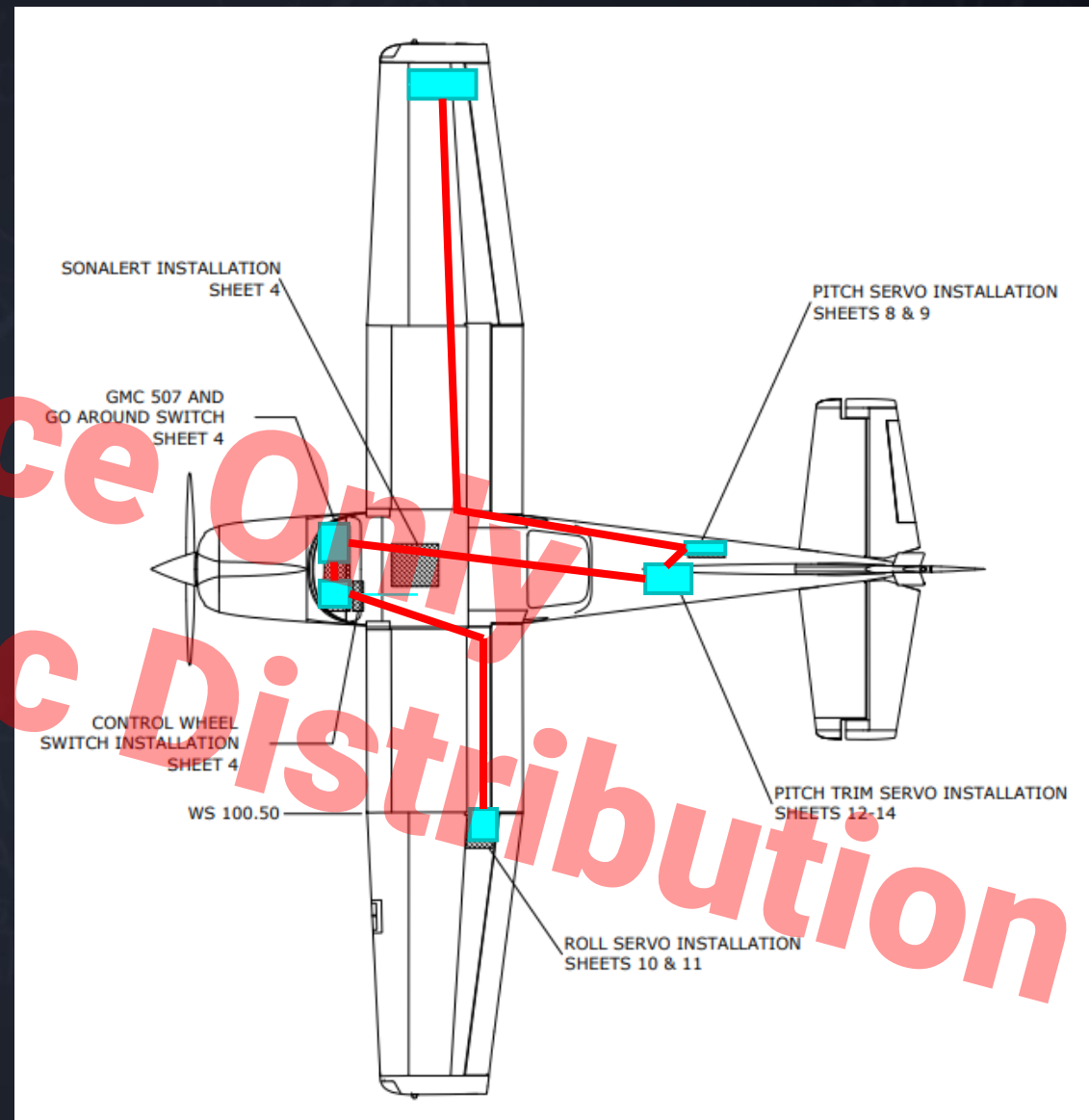
Where to put GMU 11?

- Right wing isolates it from roll servo and landing light

How well do you think this CAN bus run will work?

Problems with this layout:

- Long Length
- Backbone doubles back on itself
- Both can cause network errors

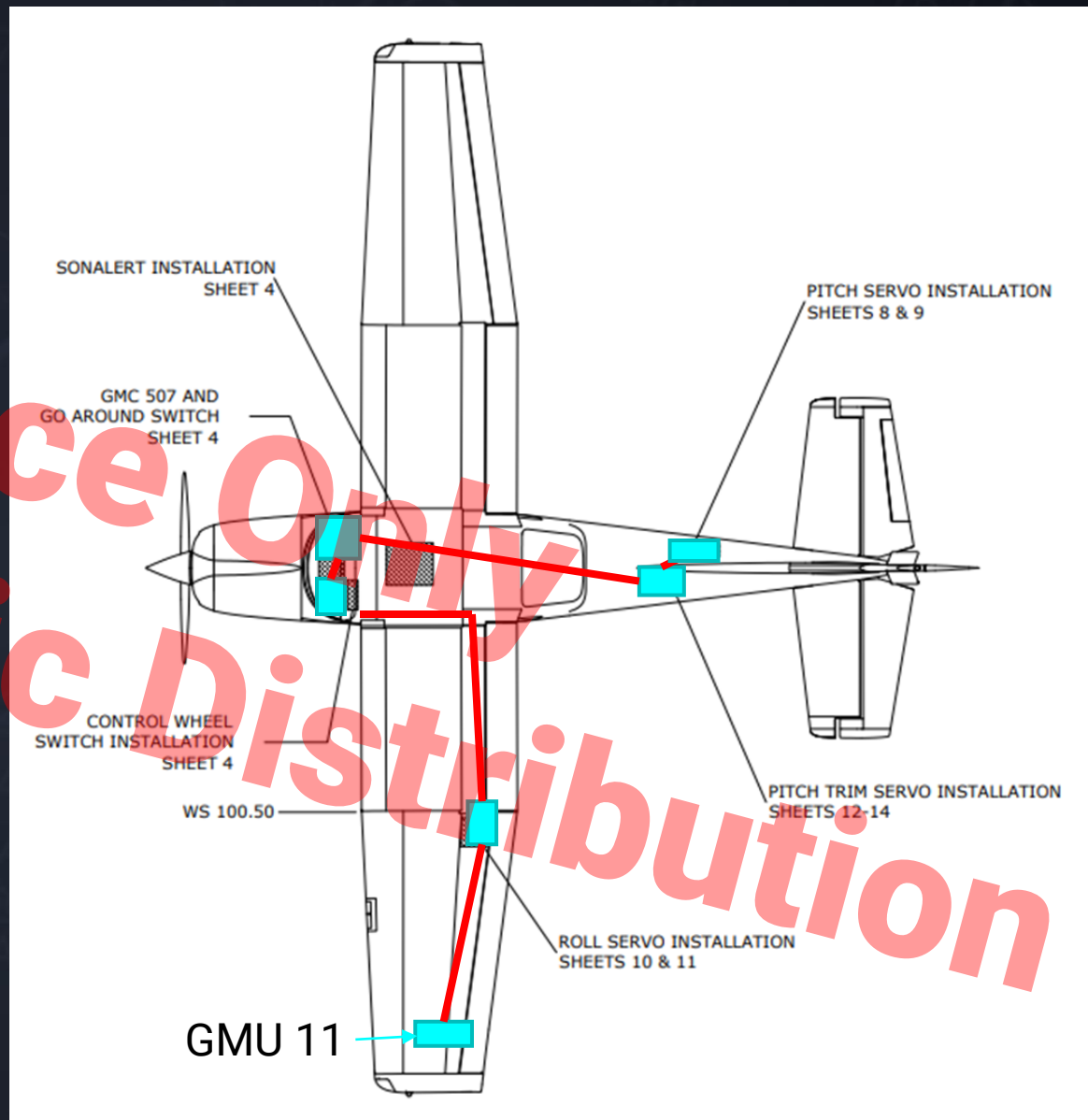


Solution: Move the GMU to the Left Wing

- ✈️ Terminate bus at GMU and Pitch Servo
- ✈️ Cuts the length nearly in half
- ✈️ Reduces/eliminates double runs of CAN bus backbone

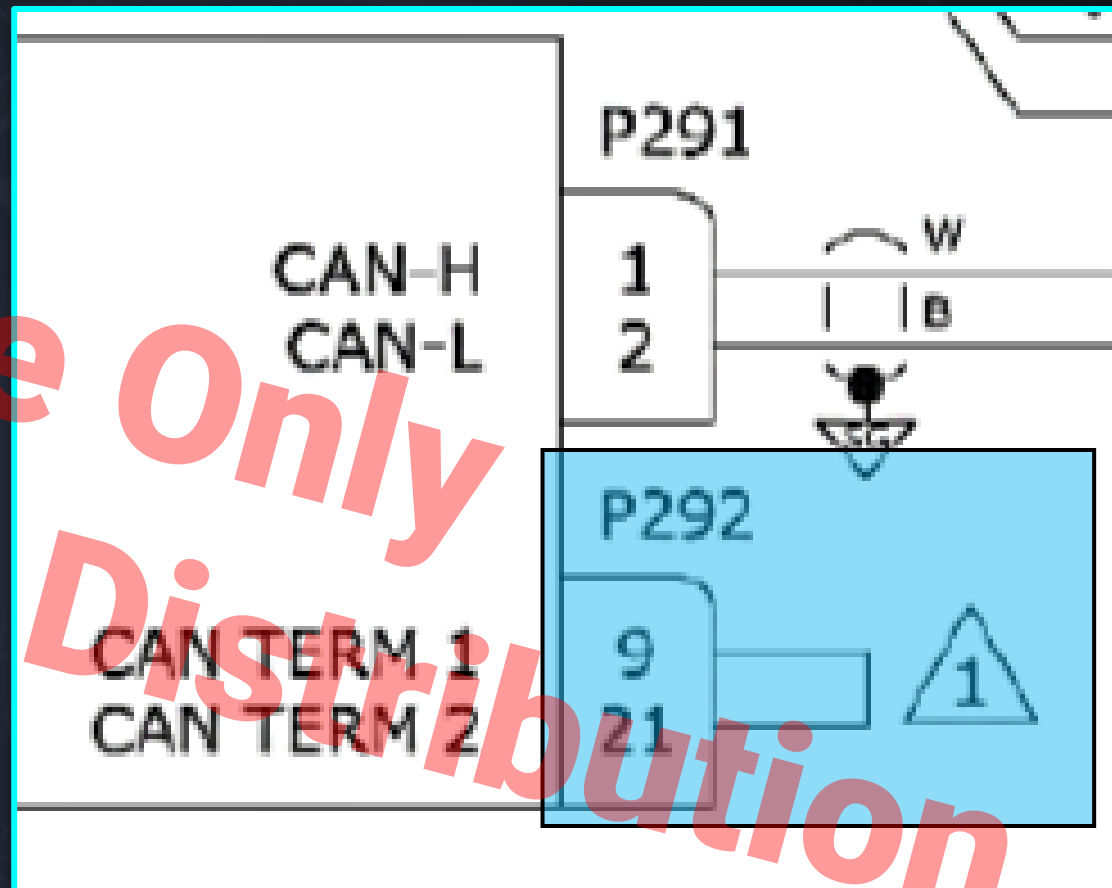
What about GMU interference?

- ✈️ Roll servo
- ✈️ Landing light
- ✈️ Appendix F: Magnetic Interference Survey

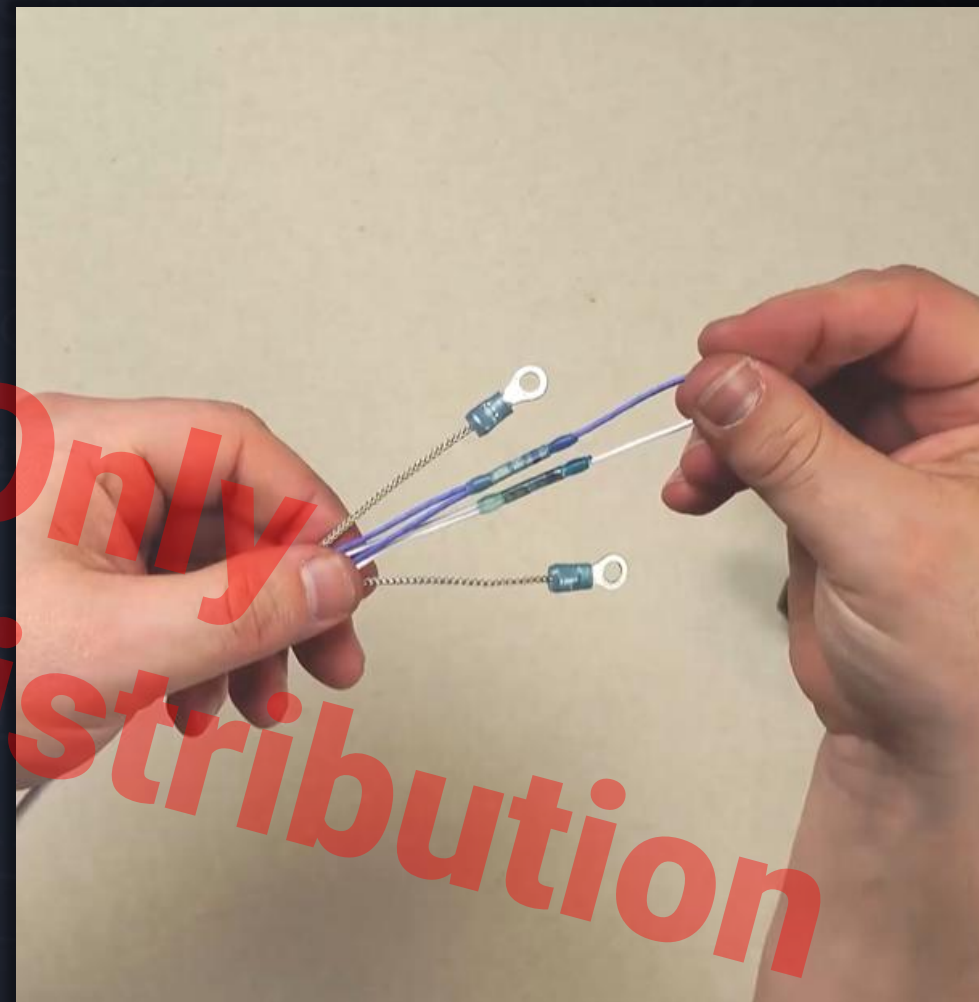


Internal CAN Termination Resistor

- Internal resistor in LRU
- Install jumper to connect CAN TERM pins
 - ✚ GSA
- OR- connect CAN LO to designated TERM pin
 - ✚ Follow the CAN bus interconnect drawing



- ✈️ Forget an LRU? Adding another one?
- ✈️ If you need to splice in additional length:
 - ✈️ Leave as little unshielded wire as possible
 - ✈️ ALWAYS connect the shields!
 - ✈️ Only use a connector if you need to. Crimp splice required.
 - ✈️ Connectors introduce a risk of signal loss due to corrosion, etc.



✈️ The TXi uses High Speed data bus

✈️ It is a high bandwidth network

✈️ This makes it incredibly robust but decisions should be made with an emphasis on data priority



✈ HSDB provides the following key features:

✈ 1. Point-to-Point Communication:

- ✈ Exploits the Ethernet hardware while eliminating the weaknesses associated with the Ethernet protocol. HSDB has only one transmitter coupled with only one receiver on each pair of wires. This avoids various concerns related to data collisions and an inadvertent constant transmitter.

✈ 2. Cyclic Redundancy Check (CRC):

- ✈ Used on data packets to ensure data integrity. CRC is used at both the IEEE 802.3 layer and the Garmin HSDB layer.

✈ 3. Guaranteed Delivery:

- ✈ Guaranteed delivery of asynchronous packets is ensured through an acknowledge protocol with multiple retries.

✈ 4. Data Age Monitoring:

- ✈ HSDB packets age is continuously monitored to ensure that old packets are flagged.

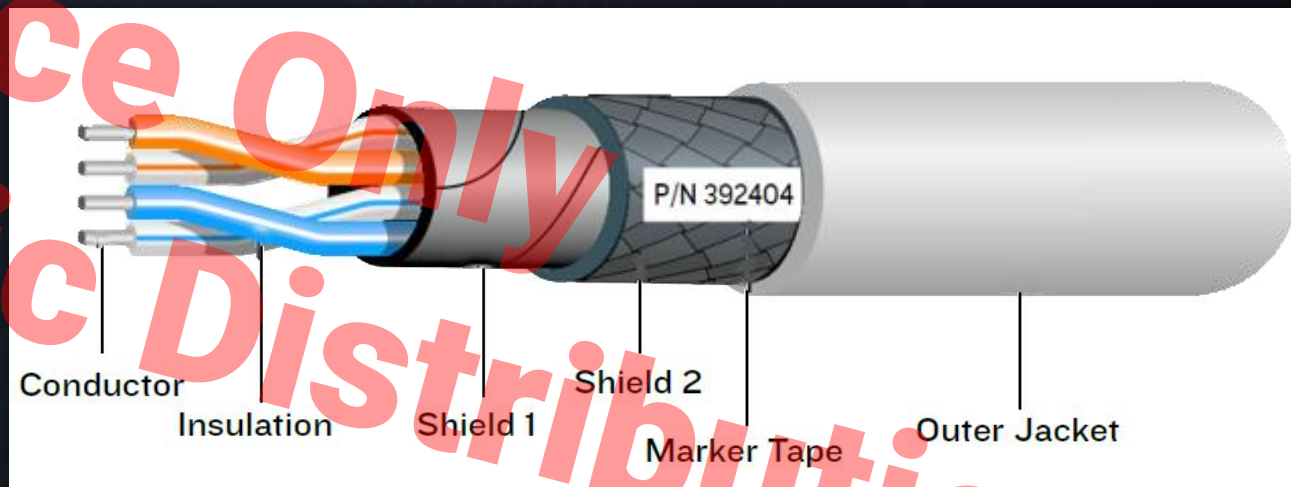
✈ 5. Routing Optimization:

- ✈ HSDB contains routing information the receiving subsystem will utilize to identify whether the data needs to be passed along to another subsystem. Routing path is optimized for system architecture.

✈ 6. Health Monitoring:

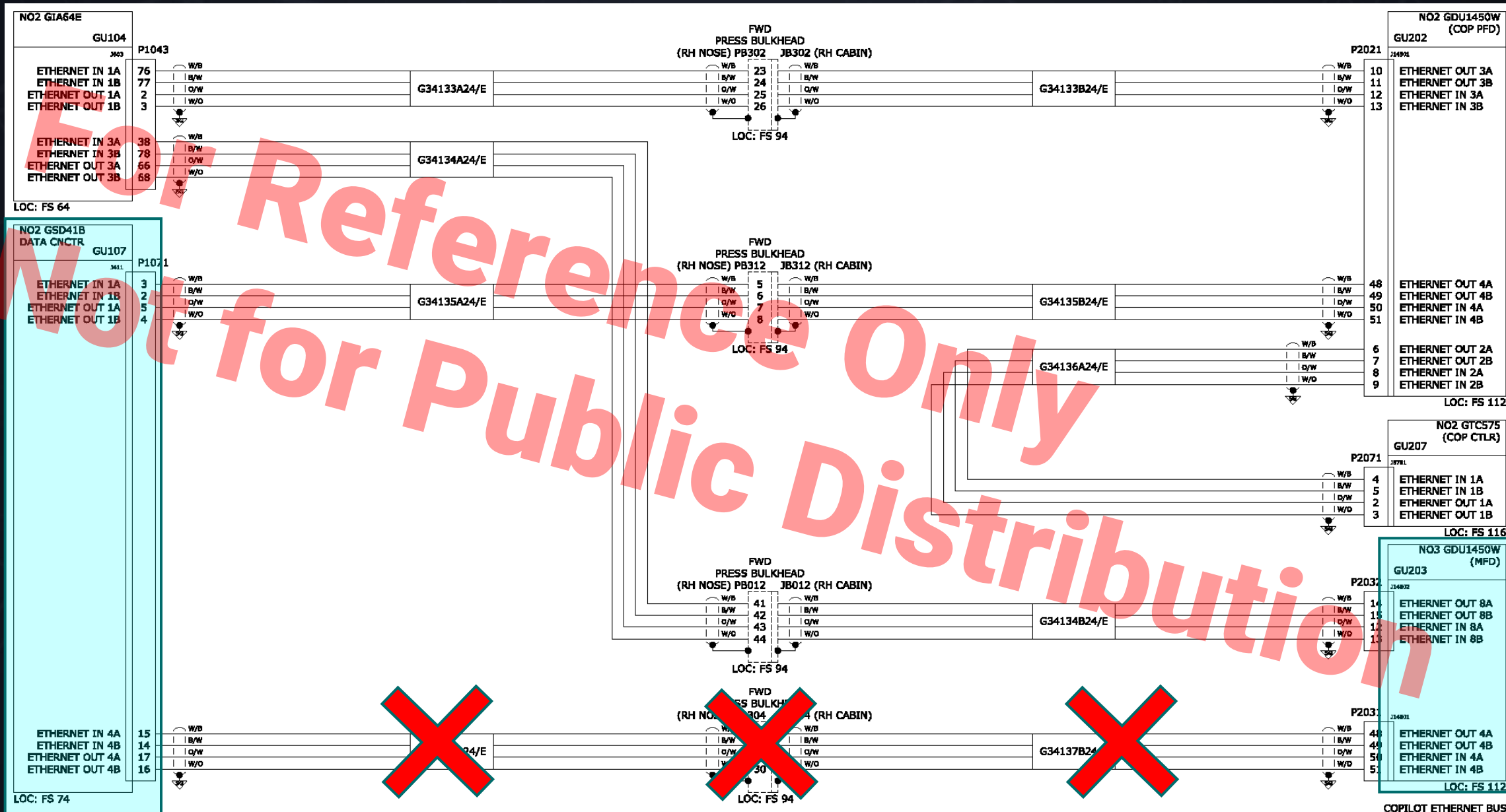
- ✈ The HSDB protocol entails continually monitoring the health of communication between all subsystems connected together. Communication failures are reported

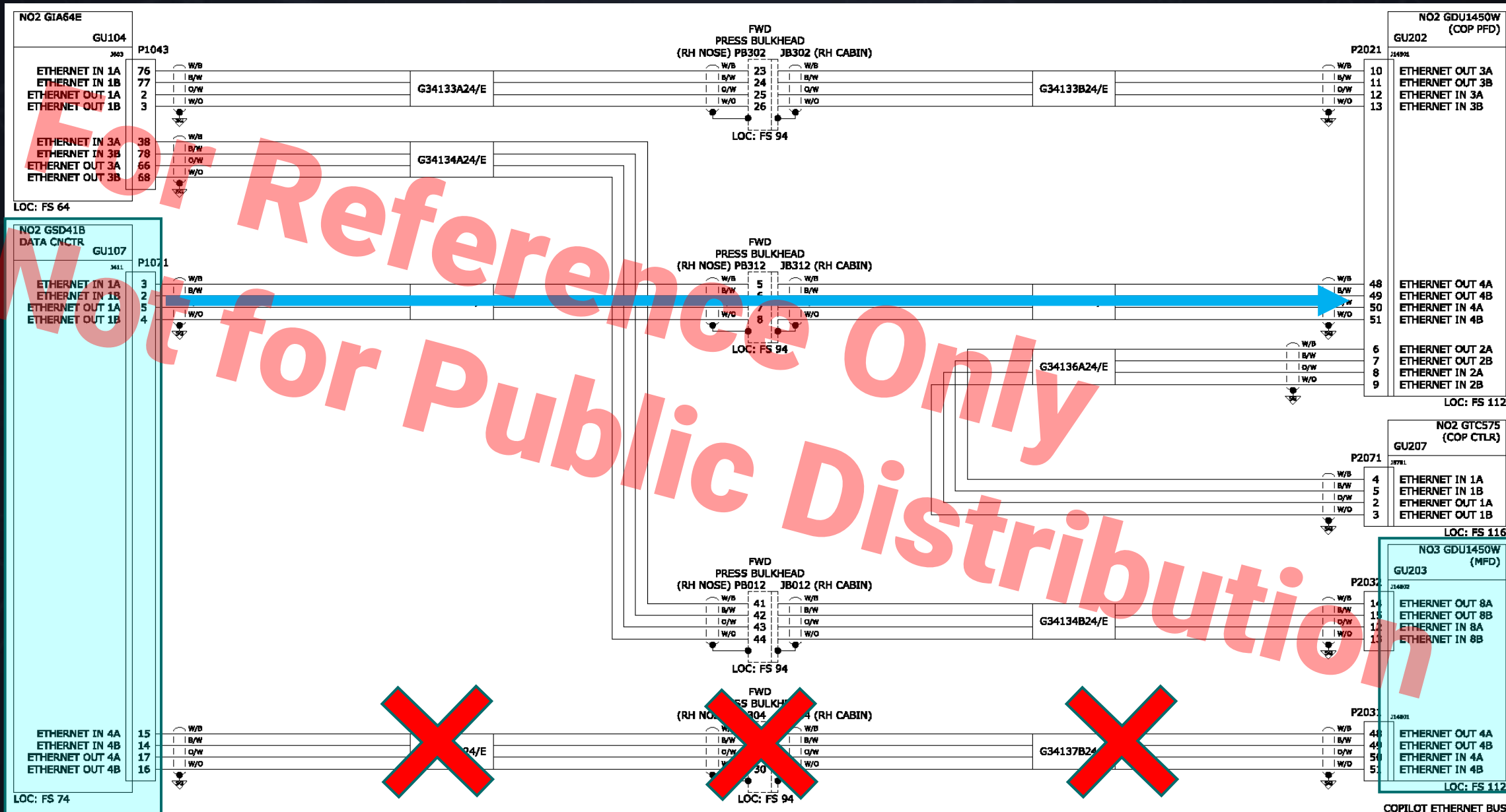
- ✈ Aluminum + Mylar foil, 100 ohms nominal impedance
- ✈ Simple 4-conductor shielded cable
- ✈ Point-to-point full-duplex communication
- ✈ Self-configuring ports



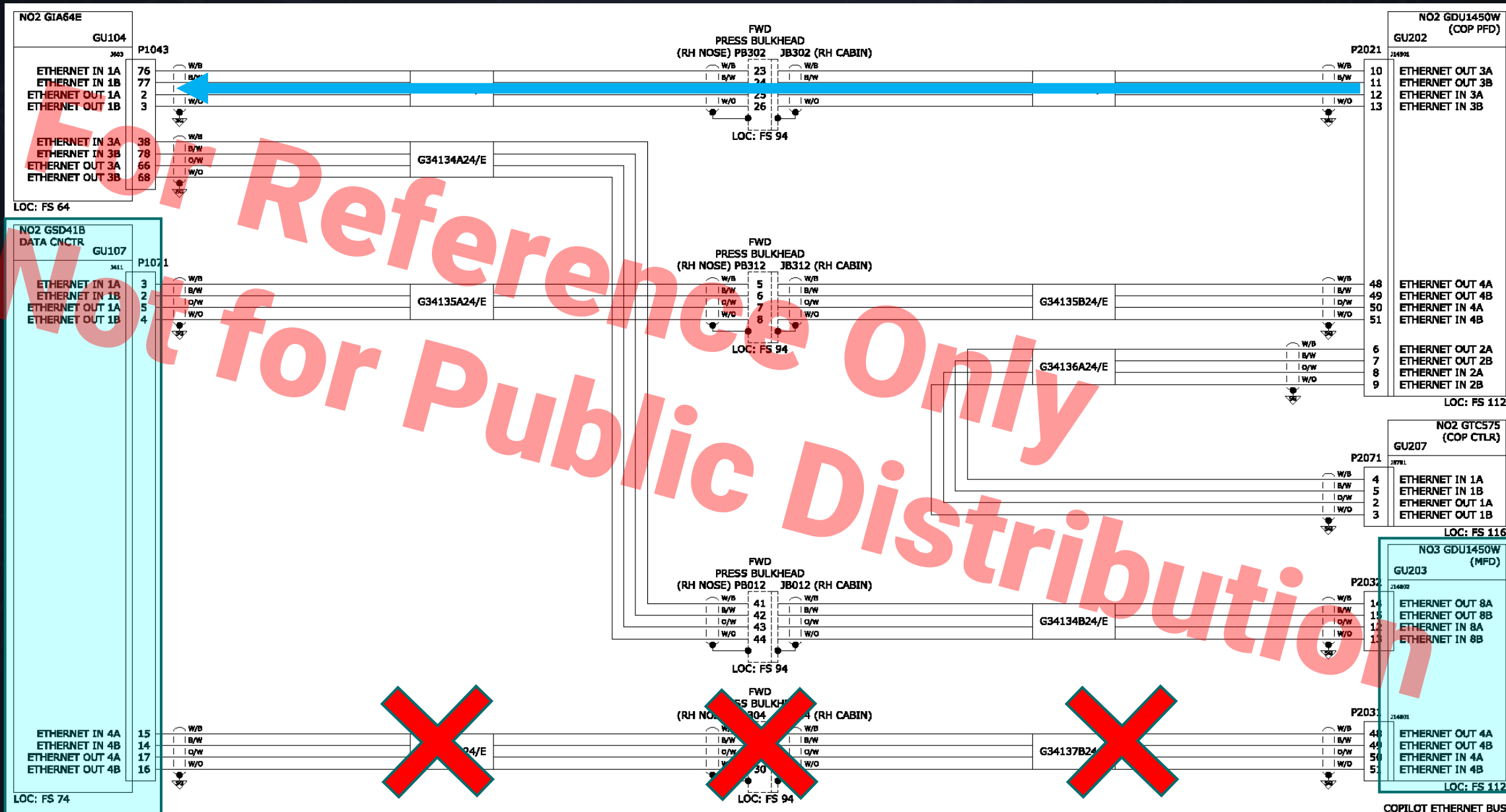


HOW Do I get DATA From GDU To GSD?

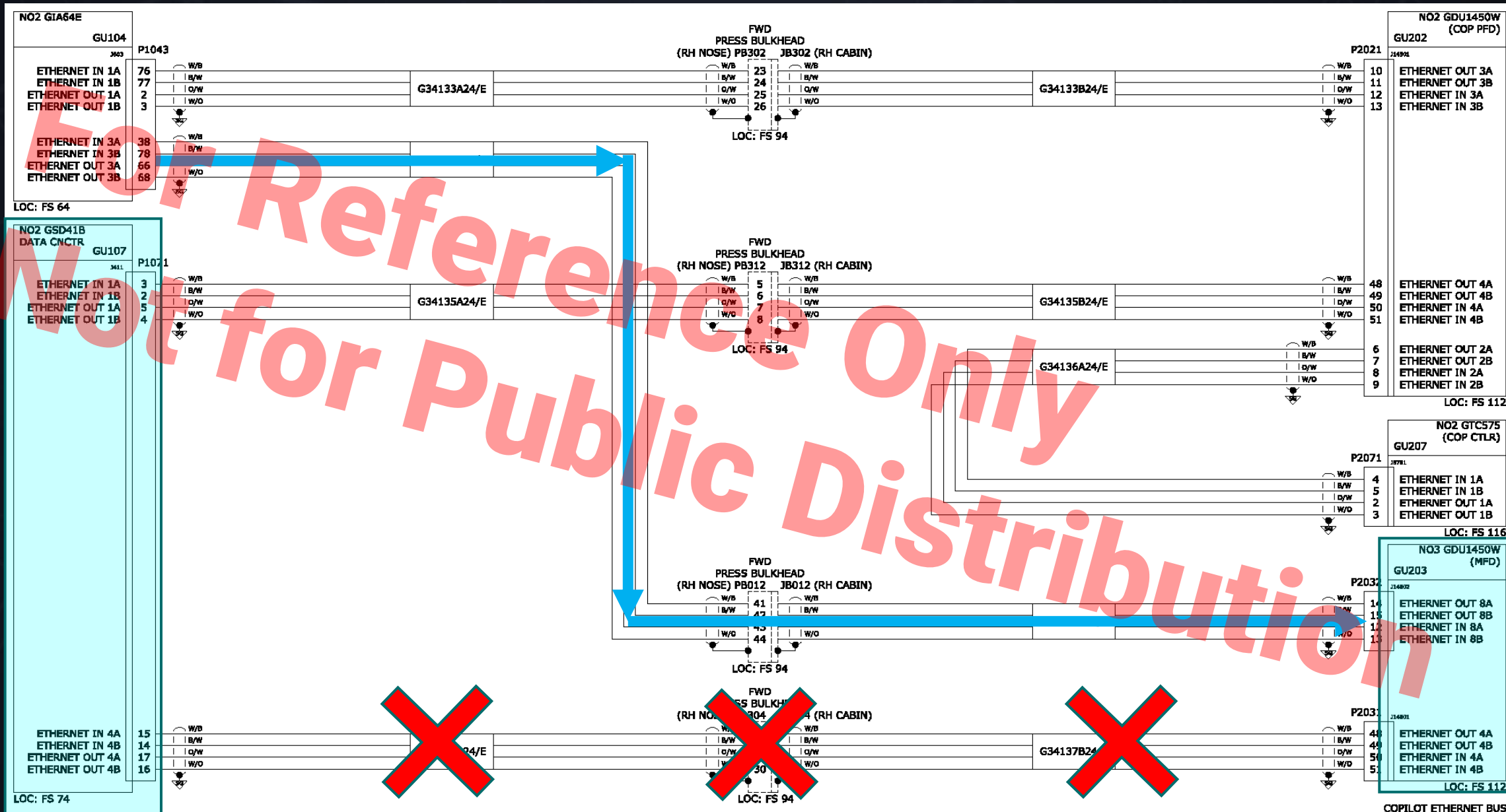




HOW Do I get DATA From GDU To GSD?



HOW Do I get DATA From GDU To GSD?





3.2.7 HSDB Architecture

The HSDB architecture allows for many connection possibilities. The Ethernet architecture limitations/options shown in this section are used as a guide for common LRU combinations. Block diagrams are illustrated in Figure 3-3 through Figure 3-14. A summary of the Garmin LRUs with HSDB capability and available ports is shown in Table 3-21.

Table 3-21 Garmin LRU HSDB Port Summary

LRU	HSDB Ports	LRU	HSDB Ports
GDU 620	2	GTX 345	2
GDU 700/1060/1210	4	GPS 175/GNX 375/GNC 355	1
GTN 6XX/7XX/Xi	4	GI 275	2
GTS 8XX	1	GDL 60	2
GDL 69 Series	4	GNC 215	1
GWX 68/70	1	GDM 450R/4500	1
GDL 88	2		

The following steps/figures are used as a guide in making HSDB connections (not all possibilities are included):

1. Pilot's PFD and the display with the second ADAHRS (if applicable) must be directly connected.
2. The TXi PFD must be directly connected to its GI 275 standby.
3. When multiple GDUs are installed, they must be connected to each other either directly or through another GDU.

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✈ There is a priority on how we connect a TXi system together

✈ Generally a unit with 4 ports is a great “hub” device and can pass data well

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The following steps/figures are used as a guide in making HSDB connections (not all possibilities are included):

1. Pilot's PFD and the display with the second ADAHRS (if applicable) must be directly connected.
2. The TXi PFD must be directly connected to its GI 275 standby.
3. When multiple GDUs are installed, they must be connected to each other either directly or through another GDU.
4. Systems with a single display and dual GTN 6XX/7XX/Xi units must connect each GTN 6XX/7XX/Xi to the display.
5. Connect other Garmin LRUs via HSDB to the pilot's PFD first, then to the display with a second ADAHRS/AHRS/ADC, if applicable. The next priority is an EIS display, then finally an MFD.
6. LRUs not installed under this STC must still meet the installation requirements that are applicable to those LRUs.
7. Choose the figure that most closely represents the aircraft's equipment and cross out any LRUs not installed. Apply the rules above to complete the HSDB connections, if necessary.
8. For GDL 60 installations that include RAS capability, the GDL 60 must be directly connected to the EIS GDU that is powered on by the RAS relays. To provide Flight Data Log offloading to the Aviation Cloud, a GDL 60 must be directly interfaced to a GDU or GI 275.
9. A direct HSDB connection is required for 100Mb Ethernet fast sync between two compatible LRUs.

✈️ There is a priority on how we connect a TXi system together

✈️ In our Example you will notice the priority

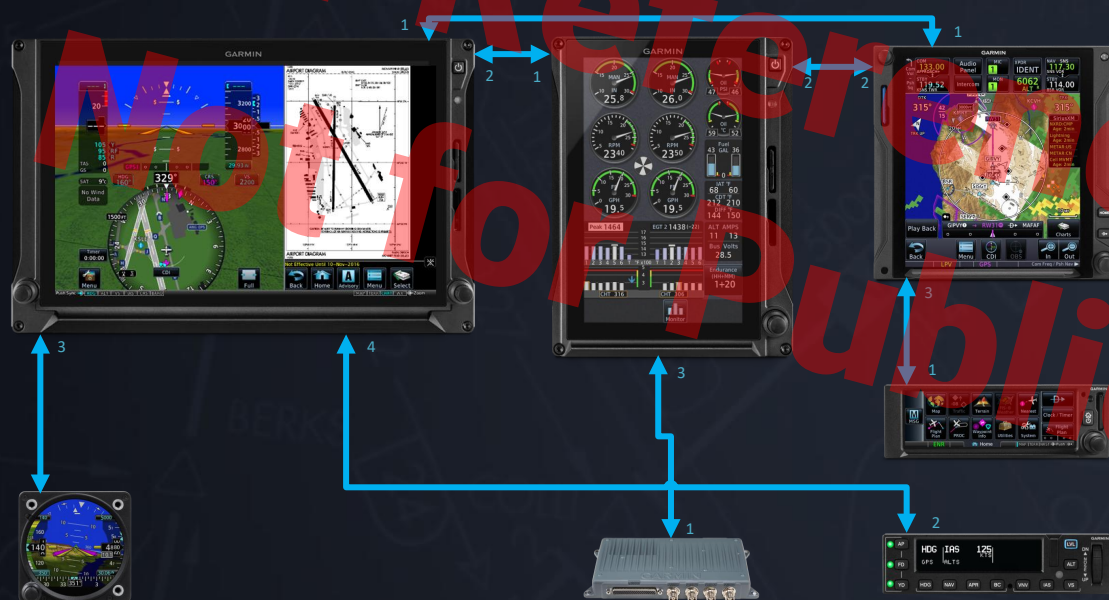
✈️ Displays direct connected

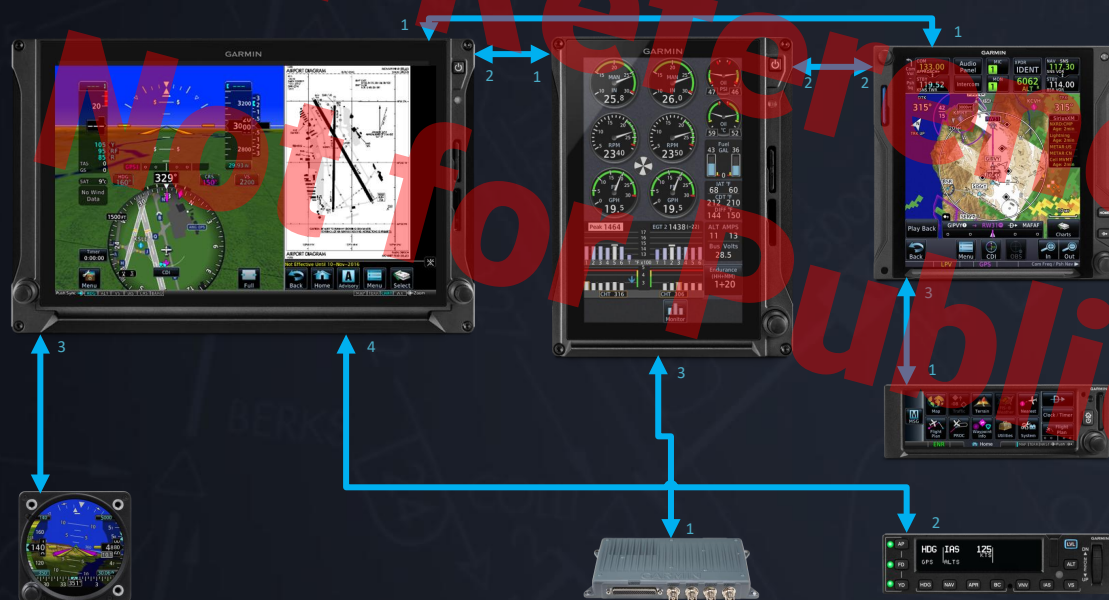
✈️ GI direct to PFD 1

✈️ GTN direct to display

✈️ GMC 605 to PFD1

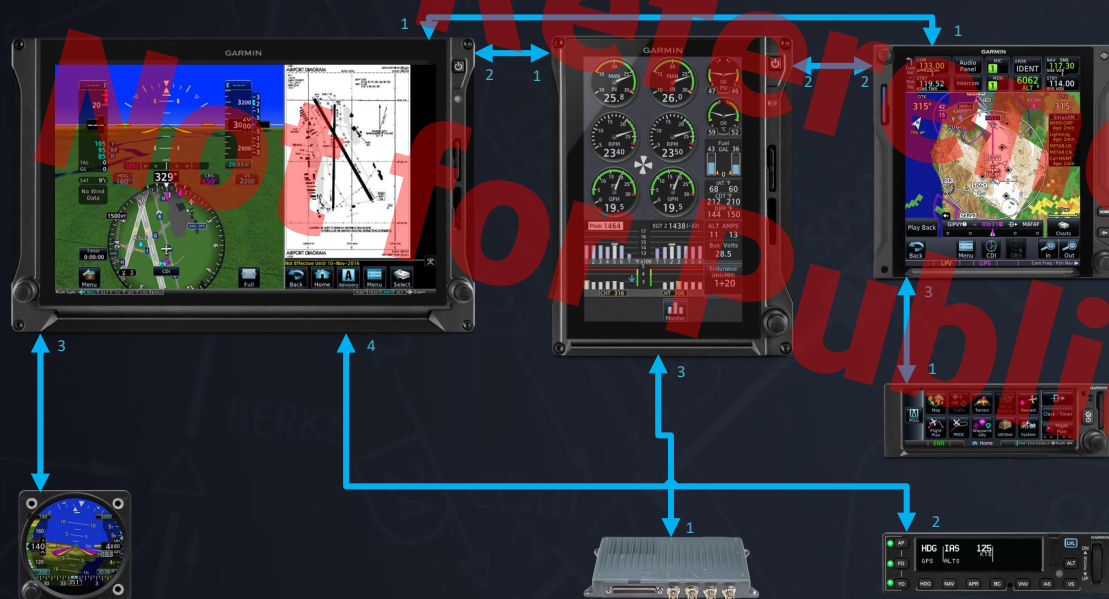
✈ Should I connect the GDL 60 to the GTNXi?





✈ Should I connect the GDL 60 to the GTNXi?

✈ Ask is it necessary for flight to require the redundancy?



✈ Should I connect the GDL 60 to the GTNXi?

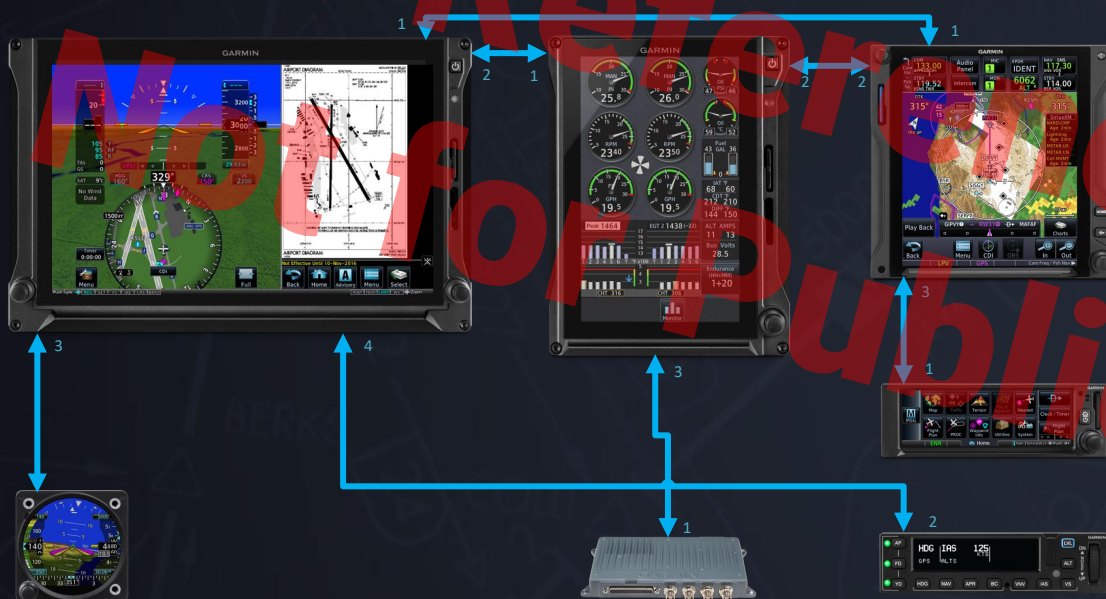
✈ Ask is it necessary for flight to require the redundancy?

✈ The answer is no.

✈ Hooking up the second HSDB port is known to cause issues for the GDL 60

✈ It can have ill effects on Flight log upload

✈️ To make sure you are making the proper decisions look at section 3 HSDB architecture for requirements



- ✈ We must plan out datapaths correctly and understand the limitations and information being passed to make the best decisions

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How many ports generally indicates an LRU passes data well?

- A. 1
- B. 2
- C. 3
- D. 4



How many ports generally indicates an LRU passes data well?

- A. 1
- B. 2
- C. 3
- D. 4



What type of splices should I use for CAN BUS

- A. Solder
- B. Connectors
- C. Environmental Crimp Splice
- D. It doesn't matter



What type of splices should I use for CAN BUS

- A. Solder
- B. Connectors
- C. Environmental Crimp Splice
- D. It doesn't matter



I attended the EFIS Design class in the Blue room.

- A. True
- B. False



I attended the EFIS Design class in the Blue room.

- A. True
- B. False

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Thank You!

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